PathFinders

PathwiseAI
Software Requirements Specifications (SRS)

Version 1.0
03/29/26

Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within <> are replaced with appropriate text.	YES
The document is prepared in the “Justify Text” format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states “The first version”. Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
The requirements are written in full sentences.	YES
The requirements are atomic, clear, unambiguous, feasible, and testable.	YES
The requirements start with “The system shall…”.	YES
The total number of “Frequency of Use” in the use cases add up to 100%.	YES
Each action in the use cases has a separate sequence number.	YES
The number of requirements is adequate to specify the details of all the features of the product.	YES
The requirements traceability matrix is complete.	YES
Each use case listed in the use case table is detailed with the use case template.	YES
All fields in the use cases are complete.	YES
In the use cases, the user actions start with “The user …” and the system actions start with “The system …”.	YES
In the use cases, the steps in each alternative scenario include the same id	YES

structure such as 2.a., 3.a., 4.a., etc.	
All requirements have correct ID numbers.	YES
The use case names start with a verb followed by a noun.	YES
All use case actions have a sequence number, and all action sequence numbers are unique.	YES
The use case scenarios include an adequate number of actions to describe the scenario.	YES
The use case covers all the necessary feature-related scenarios.	YES

Final Control	**Yes/No**
The document is checked based on the "Team Project Artifacts Common Errors" Guideline. The document is ready for release.	YES
The document is not ready for release.	NO

Document Preparation

Name	Role	Approval (Signature)	Approval Date
Meghana Nalluri	Project Requirements Manager	MN	03/29/26
Bharadwaj jakkula	Project Requirements Engineer	BJ	03/29/26
Sricharitha Katta	Project Configuration Manager	SK	03/29/26
Vineeth Reddy Kodakandla	Project Manager	VK	03/29/26

Document Approvals

Name	Role	Approval (Signature)	Approval Date
Meghana Nalluri	Project Requirements Manager	MN	03/29/26
Bharadwaj jakkula	Project Requirements Engineer	BJ	03/29/26
Sricharitha Katta	Project Configuration Manager	SK	03/29/26
Vineeth Reddy Kodakandla	Project Manager	VK	03/29/26
Meghana Nalluri	Project Requirements Manager	MN	03/29/26
Bharadwaj jakkula	Project Requirements Engineer	BJ	03/29/26
Sricharitha Katta	Project Configuration Manager	SK	03/29/26

Revision History

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Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose of the Document

The purpose of this document is to define and communicate the complete software requirements specifications for the PathWise AI system. This document identifies all functional and quality requirements that the PathWise AI platform must satisfy to meet stakeholder needs as identified in the Project Vision Document (PVD). Each requirement is written as an atomic, testable statement beginning with the phrase "The system shall" to ensure clarity and unambiguity. The document also presents detailed use cases covering all primary interaction scenarios between users and the system, and provides a requirements traceability matrix linking each requirement to its related requirements and test cases.

1.2 Scope of the Document

This Software Requirements Specification (SRS) document provides a comprehensive definition of the functional requirements, hardware requirements, quality requirements, product user descriptions, and use cases for the PathWise AI platform. PathWise AI is an intelligent Software-Defined Wide Area Network (SD-WAN) management platform designed to predict network link degradation using Long Short-Term Memory (LSTM) neural networks and to autonomously reroute mission-critical traffic before performance degradation impacts users. The document covers all five core product features as defined in the Project Vision Document: the Predictive Telemetry Engine, Autonomous Traffic Steering, Digital Twin Validation Sandbox, Intent-Based Management Interface, and the Multi-Link Health Scoreboard. Requirements related to third-party SDN controller internal implementations, physical network hardware manufacturing, and cloud hosting infrastructure provisioning are considered out of scope for this document.

1.3 References

Reference Title/Name	Date	Version
Team Pathfinders. PathWise AI – Project Vision Document (PVD)	02/15/2026	1.3
Team Pathfinders. PathWise AI – Team Project Proposal	02/03/2026	1.0
Team Pathfinders. PathWise AI – Project Plan (PP)	03/01/2026	1.0
IEEE/ISO/IEC 29148-2018 - ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle processes -- Requirements engineering	10/23/2018	2018
Web Content Accessibility Guidelines (WCAG)	11/11/2008	2.0
Hochreiter, S., & Schmidhuber, J. Long Short-Term Memory. Neural Computation.	1997	N/A
Open Networking Foundation (ONF). Software-Defined Networking (SDN) Definition. ONF White Paper.	2016	N/A
OpenDaylight Project. OpenDaylight SDN Controller Documentation. https://www.opendaylight.org	2024	N/A

ONOS Project. Open Network Operating System Documentation. https://onosproject.org	2024	N/A
Batfish Team. Batfish: A Network Configuration Analysis Tool. https://batfish.org	2024	N/A
Clemm, A. et al. Intent-Based Networking – Concepts and Definitions. IETF Draft.	2017	05
Cisco Systems. SD-WAN End-to-End Deployment Guide. Cisco Documentation.	2024	N/A

1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Software-Defined Wide Area Network	SD-WAN	A virtual WAN architecture that allows enterprises to leverage any combination of transport services to securely connect users to applications.
Software-Defined Networking	SDN	An approach to networking that uses software-based controllers or APIs to direct traffic on the network and communicate with the underlying hardware.
Long Short-Term Memory	LSTM	A type of recurrent neural network architecture designed to learn long-term dependencies in sequential data, used for time-series prediction.
Intent-Based Networking	IBM	A form of network administration that incorporates machine learning and orchestration to automate administrative tasks based on high-level policy intent.
Digital Twin	DT	A virtual representation of a physical network system used for simulation, testing, and validation of routing changes before production deployment.
Voice over Internet Protocol	VoIP	A technology that allows voice calls to be made over the internet, representing a class of mission-critical real-time application traffic.
Small-to-Medium Enterprise	SME	Organizations with fewer than 500 employees and revenue typically under $50 million, representing the primary target market for PathWise AI.
Application Programming Interface	API	A set of protocols and tools for building software applications that enables integration between system components.
Command Line Interface	CLI	A text-based interface used to interact with software and operating systems, representing the traditional complex management approach PathWise AI replaces.
Mean Squared Error	MSE	A statistical measure used to evaluate the accuracy of the LSTM prediction model by calculating the average squared difference between predicted and actual values.
YANG/NETCONF	N/A	A data modeling language (YANG) and network management protocol (NETCONF) used for configuring network devices via SDN controllers.
Hitless Handoff	N/A	A traffic steering technique that transfers active sessions between network links without any packet loss or service

		interruption.
Brownout	N/A	A condition where network link performance degrades significantly below acceptable thresholds without complete link failure, representing a predictable precursor to failure.
OpenDaylight	ODL	An open-source SDN controller platform used by PathWise AI for programmatic network flow table management.
Open Network Operating System	ONOS	An open-source SDN controller designed for high-availability production environments, supported as an alternative SDN controller integration target.
Mininet	N/A	An open-source network emulation framework used by PathWise AI to implement the Digital Twin Validation Sandbox.
Batfish	N/A	An open-source network configuration analysis and verification tool used to validate routing changes for policy compliance before production deployment.
TimescaleDB	N/A	A time-series database used by PathWise AI to store and query high-frequency network telemetry data with temporal indexing.
Simple Network Management Protocol	SNMP	A standard protocol for collecting and organizing information about managed devices on IP networks, used as a telemetry ingestion method.
NetFlow	N/A	A network protocol developed by Cisco for collecting IP traffic information and monitoring network traffic flow, used as an alternative telemetry source.

2. Product Scope

PathWise AI is an intelligent, vendor-agnostic SD-WAN management platform designed to transform enterprise network management from a reactive process into a predictive, autonomous system. The platform addresses the critical "switching gap" in traditional networking, where performance degrades or connections drop before backup links are engaged, by utilizing deep learning-based prediction to forecast link brownouts 30 to 60 seconds before they occur and to execute hitless traffic handoffs with zero packet loss.

The scope of PathWise AI encompasses five integrated functional areas. The first is a Predictive Telemetry Engine that continuously ingests real-time WAN link metrics and uses a trained LSTM neural network to generate predictive health scores for every monitored link. The second is an Autonomous Traffic Steering capability that integrates directly with SDN controllers to programmatically reroute mission-critical traffic before predicted degradation impacts users. The third is a Digital Twin Validation Sandbox that mirrors the live network using Mininet and Batfish to validate every AI-driven routing change for safety and policy compliance before production deployment. The fourth is an Intent-Based Management Interface that allows generalist IT staff to manage complex network policies using natural language commands without requiring CLI expertise. The fifth is a Multi-Link Health Scoreboard that provides real-time visualization of link health scores, AI decision confidence levels, and routing event transparency.

PathWise AI is explicitly out of scope for the physical manufacture or provisioning of network hardware, the internal implementation of SDN controllers, cloud hosting infrastructure management, and the development of physical WAN link technology. The system operates as an intelligent management overlay on top of existing network infrastructure.

3. Product Users

3.1 Users

The following table presents all identified user types for the PathWise AI platform, along with the characteristics of each user type that are relevant to system design and interface requirements.

Product User Type	User Characteristics
Network Administrator	The Network Administrator is the primary technical user who monitors and manages the SD-WAN system on a daily basis. The Network Administrator holds specialized networking knowledge and typically possesses certifications such as CCNA or equivalent industry credentials. This user is a technically proficient adult professional employed within an SME, managed service provider, or enterprise IT department. The Network Administrator directly interacts with the PathWise AI platform to review telemetry data, validate AI-driven routing decisions, and configure traffic steering policies.
Generalist IT Staff	The Generalist IT Staff is an SME IT professional who manages network policies using the simplified Intent-Based Networking (IBN) interface without requiring specialized networking certifications. This user type typically holds a general IT background and may not possess advanced networking expertise such as CCNP or JNCIS. The Generalist IT Staff relies on the natural language policy management capabilities of PathWise AI to perform network administration tasks, making them the primary beneficiary of the IBN interface design.
MSP Technician	The MSP Technician is a service provider staff member responsible for deploying and managing PathWise AI across multiple client network environments. This user type is technically proficient and manages service-level agreements (SLAs) across multiple organizational accounts simultaneously. The MSP Technician monitors predictive alerts, generates performance reports, and coordinates network policy changes with clients. This user requires access to multi-site management capabilities and centralized monitoring dashboards.
IT Manager / Business Leader	The IT Manager or Business Leader is a decision-making user who oversees network strategy, reviews AI transparency reports, and approves high-level network configurations. This user type possesses a management and business-oriented perspective rather than deep technical networking knowledge. The IT Manager evaluates system return on investment, monitors overall network performance metrics, and ensures that AI-driven decisions align with organizational business objectives and compliance requirements.
End User (Employee)	The End User is an indirect user who benefits from improved network reliability for daily operations without directly interacting with the PathWise AI management platform. The End User includes employees across various departments who rely on stable connectivity for real-time applications including VoIP, video conferencing, and cloud-based platforms. The End User may report connectivity issues to the IT team when experienced but does not access the PathWise AI system interface directly.

4. Use Cases

The following table presents the essential use cases for the PathWise AI platform. Each use case is identified by a unique Use Case ID and is elaborated in full detail in the corresponding Appendix B section of this document. The frequency of use values represent estimated percentages of all system interactions and collectively sum to 100 percent.

Use Case ID	Use Case Name	Brief Description
UC-1	Monitor Network Telemetry	This use case describes the process by which a Network Administrator or MSP Technician accesses the PathWise AI dashboard to monitor real-time network telemetry data, including latency, jitter, and packet loss metrics across all active WAN links. The use case covers the continuous display and retrieval of telemetry data from the TimescaleDB data store.
UC-2	Predict Link Degradation	This use case describes the automated process by which the PathWise AI Predictive Telemetry Engine analyzes incoming network telemetry streams using the LSTM neural network model to generate predictive health scores for each monitored WAN link and alert the Network Administrator when a link is forecast to degrade within a 30 to 60 second window.
UC-3	Execute Traffic Steering	This use case describes the process by which the PathWise AI system autonomously executes a hitless handoff of high-priority network traffic from a predicted degrading link to a stable alternative link, utilizing SDN controller integration to modify flow tables without administrator intervention and without packet loss.
UC-4	Validate Routing Change	This use case describes the process by which the PathWise AI Digital Twin Validation Sandbox automatically simulates and validates every AI-generated routing change within a Mininet-based virtual environment using Batfish policy analysis before any change is deployed to the live production network.
UC-5	Manage Network Policy	This use case describes the process by which a Generalist IT Staff member or Network Administrator accesses the Intent-Based Management Interface to define, update, or remove traffic management policies using natural language commands, which the system translates into YANG/NETCONF configurations for SDN controller deployment.
UC-6	Authenticate User	This use case describes the process by which any user type accesses the PathWise AI management platform by providing valid credentials, and the system authenticates the user and grants role-based access permissions appropriate to the user's designated role within the organization.

5. Functional Requirements

This section defines all functional requirements for the PathWise AI platform. Each requirement is written as an atomic, unambiguous statement beginning with "The system shall" and is assigned a unique identifier and a priority level of High, Medium, or Low. High priority requirements represent mandatory capabilities without which the system cannot fulfill its core value proposition. Medium priority requirements represent important capabilities that significantly enhance the system. Low priority requirements represent capabilities that improve the overall product but may be deferred to later development iterations.

5.1 Software Requirements

The following software requirements define the functional behavior of the PathWise AI platform and its interactions with users, SDN controllers, telemetry sources, and the Digital Twin Sandbox environment.

Requirement ID	**Requirement Priority** (High/Medium/Low)	**Requirement**
Req-Func-Sw-1	High	The system shall ingest real-time network telemetry metrics, including latency, jitter, and packet loss, from all monitored WAN links at a minimum polling frequency of once per second.
Req-Func-Sw-2	High	The system shall utilize a trained LSTM neural network model to forecast WAN link performance degradation for a prediction window of 30 to 60 seconds into the future.
Req-Func-Sw-3	High	The system shall generate a predictive health score between 0 and 100 for each monitored WAN link based on the real-time and historical telemetry data ingested from that link.
Req-Func-Sw-4	High	The system shall trigger an autonomous traffic steering action when the predictive health score of any active WAN link falls below a configurable administrator-defined threshold value.
Req-Func-Sw-5	High	The system shall integrate with SDN controllers, specifically OpenDaylight and ONOS, using their respective northbound APIs to dynamically read and modify network flow tables.
Req-Func-Sw-6	High	The system shall execute a hitless handoff by pre-emptively rerouting high-priority traffic classes, including VoIP, video conferencing, and financial transactions, from a degrading WAN link to the most stable available alternative link before degradation impacts user sessions.
Req-Func-Sw-7	High	The system shall maintain all active session states during traffic steering transitions to ensure no application session is interrupted or terminated during a link handoff.
Req-Func-Sw-8	High	The system shall automatically submit every AI-generated routing change to the Digital Twin Validation Sandbox for simulation and policy verification prior to any production network deployment.
Req-Func-Sw-9	High	The system shall use the Mininet network emulation framework to construct a virtual replica of the live network topology within the Digital Twin Sandbox for each validation request.
Req-Func-Sw-10	High	The system shall use Batfish to analyze every proposed routing change for routing loop freedom and compliance with existing firewall and security policies before approving the change for production deployment.
Req-Func-Sw-11	High	The system shall provide an Intent-Based Networking (IBN) interface that accepts natural language policy commands from administrators without requiring any knowledge of CLI syntax.

Req-Func-Sw-12	High	The system shall translate administrator-provided natural language policy commands into valid YANG/NETCONF configuration payloads and submit them to the integrated SDN controller for deployment.
Req-Func-Sw-13	Medium	The system shall display a real-time Multi-Link Health Scoreboard that presents the current predictive health score for every available WAN path, including Fiber, Satellite, 5G, and Broadband connections.
Req-Func-Sw-14	Medium	The system shall display the LSTM prediction confidence level and a human-readable explanation of the reasoning behind every automated path switch decision on the Multi-Link Health Scoreboard.
Req-Func-Sw-15	High	The system shall provide role-based access control that enforces distinct access permissions for each user type, including Network Administrator, IT Manager, MSP Technician, Generalist IT Staff, and End User.
Req-Func-Sw-16	High	The system shall authenticate all users using a secure credential-based login mechanism, with all credentials stored using one-way cryptographic hashing algorithms.
Req-Func-Sw-17	Medium	The system shall send real-time alert notifications to Network Administrators via the dashboard and email when a predictive health score for any monitored WAN link drops below the configured threshold value.
Req-Func-Sw-18	Medium	The system shall maintain a persistent, tamper-evident audit log recording all AI-driven routing decisions, including the prediction health scores, confidence levels, Digital Twin validation results, and routing changes applied.
Req-Func-Sw-19	Medium	The system shall support the concurrent monitoring and management of at least 100 network sites with multiple WAN links each from a single centralized management dashboard.
Req-Func-Sw-20	Low	The system shall integrate with existing network monitoring infrastructure by accepting telemetry data from devices via SNMP version 2c or higher and NetFlow version 9 or higher protocols.
Req-Func-Sw-21	Low	The system shall provide exportable performance reports for each managed network site, including historical health scores, prediction accuracy metrics, and traffic steering event logs, in PDF and CSV formats.

5.2 Hardware Requirements

The following hardware requirements define the minimum infrastructure specifications necessary for deploying and operating the PathWise AI platform.

Requirement ID	**Requirement Priority (High/Medium/Low)**	**Requirement**
Req-Func-Hw-1	High	The system shall operate on server hardware with a minimum of 32GB RAM to support concurrent LSTM model inference, SDN

		controller API communication, and Digital Twin simulation operations
Req-Func-Hw-2	High	The system shall require all managed network devices to support an SDN-capable protocol, specifically OpenFlow version 1.3 or higher, to enable programmatic flow table modification for autonomous traffic steering.
Req-Func-Hw-3	High	The system shall be deployable on commodity x86-64 server hardware and shall not require any proprietary vendor-specific networking hardware appliances.
Req-Func-Hw-4	Medium	The system shall require a minimum of 1TB of SSD storage capacity on the server platform to support TimescaleDB telemetry data retention and LSTM model training dataset storage.
Req-Func-Hw-5	High	The system shall require a minimum dedicated management network connection of 100 Mbps for real-time telemetry ingestion from all monitored WAN links.
Req-Func-Hw-6	Medium	The system shall require a server processor with a minimum of 8 CPU cores to handle concurrent telemetry stream processing, LSTM inference, and Digital Twin simulation workloads.
Req-Func-Hw-7	Low	The system shall support deployment in virtualized environments, including VMware ESXi and KVM hypervisors, to enable cloud-hosted and on-premises SaaS deployment models.
Req-Func-Hw-8	Low	The system shall support a display monitor with a minimum resolution of 1920x1080 pixels on Network Administrator workstations for optimal rendering of the Multi-Link Health Scoreboard visualization.

6. Quality Requirements

The following quality requirements define the non-functional performance, reliability, security, usability, and scalability targets that the PathWise AI platform must achieve. These requirements are derived from the Quality Ranges section of the Project Vision Document and are expressed as measurable, testable specifications.

Requirement ID	**Requirement Priority (High/Medium/Low)**	**Requirement**
Req-Qual-Perf-1	High	The system shall achieve a minimum of 90% prediction accuracy, measured by Mean Squared Error against ground truth telemetry values, for 30 to 60 second forecasts of WAN link latency, jitter, and packet loss.
Req-Qual-Perf-2	High	The system shall execute end-to-end traffic steering decisions, from predictive trigger to SDN flow table modification, within 50 milliseconds to achieve hitless handoff without packet loss.
Req-Qual-Perf-3	High	The system shall complete the full Digital Twin Sandbox validation cycle, including Mininet topology construction and Batfish policy analysis, within 5 seconds for each proposed routing change.
Req-Qual-Perf-4	Medium	The system shall respond to all IBN dashboard user interface operations, including natural language policy query processing, within 2 seconds under normal operating load conditions.
Req-Qual-Sec-1	High	The system shall encrypt all network telemetry data and management communications in transit using TLS version 1.3 or higher.

Req-Qual-Sec-2	High	The system shall encrypt all telemetry data and user credential data stored at rest using AES-256 encryption.
Req-Qual-Sec-3	Medium	The system shall comply with HIPAA data handling requirements, including comprehensive audit logging and access controls, for all deployments within healthcare facility environments.
Req-Qual-Use-1	High	The system shall enable all common network administrative tasks, including policy creation, traffic steering rule configuration, and health score monitoring, to be completed through the IBN interface without requiring any command-line interface knowledge.
Req-Qual-Rel-1	High	The system shall maintain a minimum platform availability of 99.9% on an annual basis for the PathWise AI management platform, including all core prediction, monitoring, and control services.
Req-Qual-Rel-2	High	The system shall implement automated failover and redundancy mechanisms for all management platform services to ensure continuous operation in the event of a single component failure.
Req-Qual-Rel-3	Medium	The system shall perform automated database backups of all telemetry data and configuration records at a minimum frequency of once every 24 hours and store backups in a geographically separate location.
Req-Qual-Scal-1	Medium	The system shall support the concurrent management of at least 100 network sites without measurable degradation in prediction accuracy, response time, or dashboard rendering performance.

7. Appendix A – Requirements Traceability

The following requirements traceability matrix maps each requirement identified in this document to its related requirements and the associated test case identifier that will be used to verify the requirement during the system testing phase.

Requirement ID	Related Requirements	Test Case ID
Req-Func-Sw-1	Req-Func-Hw-1 Req-Func-Hw-5 Req-Qual-Perf-1	Test-Case-1, Test-Case-2
Req-Func-Sw-2	Req-Func-Sw-1 Req-Func-Hw-1 Req-Qual-Perf-1	Test-Case-2, Test-Case-7
Req-Func-Sw-3	Req-Func-Sw-2 Req-Qual-Perf-1	Test-Case-2, Test-Case-3
Req-Func-Sw-4	Req-Func-Sw-3 Req-Func-Sw-5 Req-Qual-Perf-2	Test-Case-3, Test-Case-4
Req-Func-Sw-5	Req-Func-Hw-2 Req-Func-Sw-4 Req-Qual-Perf-2	Test-Case-4, Test-Case-5
Req-Func-Sw-6	Req-Func-Sw-4 Req-Func-Sw-5, Req-Func-Sw-7	Test-Case-5, Test-Case-6
Req-Func-Sw-7	Req-Func-Sw-6 Req-Qual-Perf-2	Test-Case-5
Req-Func-Sw-8	Req-Func-Sw-9 Req-Func-Sw-10 Req-Qual-Perf-3	Test-Case-7, Test-Case-8
Req-Func-Sw-9	Req-Func-Sw-8 Req-Func-Hw-1 Req-Qual-Perf-3	Test-Case-8
Req-Func-Sw-10	Req-Func-Sw-9 Req-Func-Sw-8	Test-Case-8, Test-Case-9
Req-Func-Sw-11	Req-Func-Sw-12 Req-Qual-Use-1 Req-Qual-Perf-4	Test-Case-10, Test-Case-11
Req-Func-Sw-12	Req-Func-Sw-11 Req-Func-Sw-5	Test-Case-11
Req-Func-Sw-13	Req-Func-Sw-3 Req-Func-Sw-14 Req-Qual-Scal-1	Test-Case-12, Test-Case-13
Req-Func-Sw-14	Req-Func-Sw-13 Req-Func-Sw-2	Test-Case-13
Req-Func-Sw-15	Req-Func-Sw-16	Test-Case-14, Test-Case-15

	Req-Qual-Sec-1	
Req-Func-Sw-16	Req-Func-Sw-15 Req-Qual-Sec-2	Test-Case-15
Req-Func-Sw-17	Req-Func-Sw-3 Req-Func-Sw-4	Test-Case-3, Test-Case-16
Req-Func-Sw-18	Req-Qual-Sec-3 Req-Func-Sw-8	Test-Case-17
Req-Func-Sw-19	Req-Func-Hw-1 Req-Qual-Scal-1	Test-Case-18
Req-Func-Sw-20	Req-Func-Sw-1 Req-Func-Hw-5	Test-Case-1
Req-Func-Sw-21	Req-Func-Sw-18	Test-Case-17
Req-Func-Hw-1	Req-Func-Sw-2 Req-Func-Sw-9 Req-Func-Sw-19	Test-Case-2, Test-Case-8, Test-Case-18
Req-Func-Hw-2	Req-Func-Sw-5 Req-Func-Sw-6	Test-Case-4, Test-Case-5
Req-Func-Hw-3	Req-Func-Hw-1	Test-Case-19
Req-Func-Hw-4	Req-Func-Sw-1 Req-Func-Sw-18	Test-Case-1, Test-Case-17
Req-Func-Hw-5	Req-Func-Sw-1 Req-Func-Sw-20	Test-Case-1
Req-Func-Hw-6	Req-Func-Hw-1 Req-Qual-Perf-2	Test-Case-19
Req-Func-Hw-7	Req-Func-Hw-3	Test-Case-20
Req-Func-Hw-8	Req-Func-Sw-13	Test-Case-12
Req-Func-Perf-1	Req-Func-Sw-2 Req-Func-Sw-3	Test-Case-2, Test-Case-7
Req-Func-Perf-2	Req-Func-Sw-6 Req-Func-Sw-5	Test-Case-5, Test-Case-6
Req-Func-Perf-3	Req-Func-Sw-8 Req-Func-Sw-9	Test-Case-8, Test-Case-9
Req-Func-Perf-4	Req-Func-Sw-11 Req-Func-Sw-13	Test-Case-10, Test-Case-12
Req-Func-Rel-1	Req-Qual-Rel-2 Req-Func-Sw-15	Test-Case-21
Req-Func-Rel-2	Req-Qual-Rel-1	Test-Case-21
Req-Func-Rel-3	Req-Func-Sw-18 Req-Func-Hw-4	Test-Case-17
Req-Func-Sec-1	Req-Func-Sw-16 Req-Qual-Sec-2	Test-Case-15, Test-Case-22
Req-Func-Sec-2	Req-Func-Sw-16 Req-Qual-Sec-1	Test-Case-15, Test-Case-22

Req-Func-Sec-3	Req-Func-Sw-18 Req-Func-Sw-15	Test-Case-17, Test-Case-14
Req-Func-Use-1	Req-Func-Sw-11 Req-Func-Sw-12	Test-Case-10, Test-Case-11
Req-Func-Scal-2	Req-Func-Sw-19 Req-Func-Hw-1	Test-Case-18

8. Appendix B – Use Case 1 – Monitor Network Telemetry

The following sections present the complete use case templates for each use case identified in Section 4 of this document. Each use case is detailed with all required fields, including the primary and secondary actors, preconditions, postconditions, trigger, main success scenario, and alternate scenarios.

Use Case ID	UC-1	
Use Case Name	Monitor Network Telemetry	
Brief Description	This use case describes the process by which a Network Administrator or MSP Technician accesses the PathWise AI dashboard to monitor real-time network telemetry data, including latency, jitter, and packet loss metrics across all active WAN links. The use case covers the continuous display and retrieval of telemetry data from the TimescaleDB data store.	
Frequency of Use	High – 30%	
Priority	High	
Current Version	1.0	
Date of First Version	03/29/2026	
Date of Current Version	03/29/2026	
Created By	Meghana Nalluri	
Last Update By	Meghana Nalluri	
Approved By	Vineeth Reddy Kodakandla	
Assumptions	The user is authenticated and logged into the PathWise AI management platform. All monitored WAN links are transmitting telemetry data to the system at the normal polling frequency. The TimescaleDB telemetry data store is operational and accessible by the system.	
Primary Actor	Network Administrator	
Secondary Actor/s	MSP Technician	
Preconditions	The user has valid PathWise AI credentials and is logged into the system. At least one WAN link is active and transmitting telemetry to the PathWise AI platform. The telemetry ingestion service is running and populating TimescaleDB.	
Postconditions	The user has reviewed real-time telemetry data for one or more WAN links. The system continues to refresh telemetry data at one-second intervals while the monitoring view is active.	
Trigger	The user navigates to the Telemetry monitoring section of the PathWise AI dashboard.	
Main Success Scenario	User Actions	System Actions
	1. The user opens a web browser and navigates to the PathWise AI management dashboard URL.	2. The system authenticates the user session and loads the management dashboard interface.
	3. The user selects the "Telemetry" view	

	from the main navigation menu. 5. The user selects a specific WAN link from the link selector panel to view its detailed metrics. 7. The user adjusts the time range filter to display telemetry data for the past 30 minutes. 8. The user reviews the real-time latency, jitter, and packet loss graphs for the selected WAN link. 9.. The user notes any anomalies or trends visible in the telemetry graphs. 10. The user selects another WAN link from the panel to compare performance across links.	4. The system queries TimescaleDB and displays the real-time telemetry monitoring view, showing all active WAN links. 6. The system retrieves and renders the historical and live telemetry data streams for the selected WAN link in graphical format.
Alternate Scenarios	User Actions	System Actions
	5.a. The user attempts to access a WAN link for a site that is offline and not transmitting telemetry. 7.b. The user selects a custom time range that exceeds the available data retention period.	6.a. The system displays a notification indicating that the selected WAN link is currently offline and no live telemetry data is available, while showing the most recent cached telemetry values. 8.b. The system displays telemetry data for the maximum available retention period and notifies the user that the requested range exceeds the retained data window.
Additional Notes (Constraints, etc.)	The telemetry monitoring view must support concurrent sessions without performance degradation for at least 10 simultaneous administrator users.	

9. Appendix B – Use Case 2 – Predict Link Degradation

Use Case ID	UC-2	
Use Case Name	Predict Link Degradation	
Brief Description	This use case describes the automated process by which the PathWise AI Predictive Telemetry Engine analyzes incoming network telemetry streams using the LSTM neural network model to generate predictive health scores for each monitored WAN link, and alerts the Network Administrator when a link is forecast to degrade within a 30 to 60 second window.	
Frequency of Use	High – 25%	
Priority	High	
Current Version	1.0	
Date of First Version	03/29/2026	
Date of Last Version	03/29/2026	
Created By	Bharadwaj Jakkula	
Last Update By	Bharadwaj Jakkula	
Approved By	Vineeth Reddy Kodakandla	
Assumptions	The telemetry ingestion service is continuously collecting data from all monitored WAN links. The LSTM model has been trained on a minimum of 30 days of historical telemetry data and is deployed to the inference service. Alert thresholds have been configured by the Network Administrator prior to the use case execution.	
Primary Actor	PathWise AI Predictive Telemetry Engine (automated)	
Secondary Actor/s	Network Administrator	
Preconditions	The telemetry ingestion service is actively receiving metrics from all monitored WAN links. The LSTM model inference service is running and operational. Alert notification channels are configured and accessible.	
Postconditions	A predictive health score has been generated and stored for each monitored WAN link. An alert notification has been sent to the Network Administrator if a health score dropped below the configured threshold. The routing recommendation has been forwarded to the traffic steering module for action.	
Trigger	The system receives a new telemetry data point from any monitored WAN link.	
Main Success Scenario	User Actions	System Actions
	3. The user (Network Administrator) reviews the health score displayed on the Multi-Link Health Scoreboard.	1. The system receives the incoming telemetry data point from the WAN link telemetry ingestion service. 2. The system appends the new data point to the LSTM model input sequence buffer for the corresponding WAN link.

		4. The system feeds the updated telemetry sequence buffer through the trained LSTM model to generate a 30 to 60 second predictive health score. 5. The system stores the generated health score in TimescaleDB with a timestamp for audit and traceability purposes. 6. The system compares the new health score against the configured alert threshold for the respective WAN link. 7. The system generates a predictive alert when the health score falls below the configured threshold, recording the confidence level and predicted degradation timeline. 8. The system sends the alert notification to all registered Network Administrator users via the dashboard notification panel and email.
	9. The user acknowledges the predictive alert notification in the dashboard.	10. The system forwards the routing change recommendation with its prediction confidence score to the Autonomous Traffic Steering module for evaluation.
Alternate Scenarios	**User Actions**	**System Actions**
	6.a. The system determines that the incoming telemetry data point is malformed or outside the valid metric range.	6.a. The system logs the malformed data point with an error classification in the audit log and discards it from the LSTM input sequence, continuing normal processing with the previous valid data point.
	7.b. The system generates consecutive alerts for the same WAN link within a five-second window.	7.b. The system suppresses duplicate alert notifications for the same link within a configurable suppression window to prevent alert flooding, while logging all generated health scores internally..
Additional Notes (Constraints, etc.)	The LSTM inference service must complete each prediction cycle within the 1-second telemetry polling interval to maintain real-time prediction accuracy.	

10. Appendix B – Use Case 3 – Execute Traffic Steering

Use Case ID	UC-3

Use Case Name	Execute Traffic Steering	
Brief Description	This use case describes the process by which the PathWise AI system autonomously executes a hitless handoff of high-priority network traffic from a predicted degrading link to a stable alternative link. The system utilizes SDN controller integration to modify flow tables without administrator intervention and without packet loss, ensuring seamless continuity of mission-critical application sessions.	
Frequency of Use	High – 20%	
Priority	High	
Current Version	1.0	
Date of First Version	03/29/2026	
Date of Last Version	03/29/2026	
Created By	Sricharitha Katta	
Last Update By	Sricharitha Katta	
Approved By	Vineeth Reddy Kodakandla	
Assumptions	The Digital Twin Validation Sandbox has successfully validated the proposed routing change before this use case initiates the production deployment. At least one stable alternative WAN link with sufficient capacity exists to absorb the rerouted traffic. The SDN controller is operational and its northbound API is accessible by the PathWise AI control interface.	
Primary Actor	PathWise AI Autonomous Traffic Steering Module (automated)	
Secondary Actor/s	Network Administrator	
Preconditions	A routing change recommendation with a passing Digital Twin validation result has been received from the validation sandbox module. The target alternative WAN link is active and has sufficient bandwidth to accommodate the rerouted traffic. The SDN controller northbound API is reachable and authenticated.	
Postconditions	High-priority traffic has been fully rerouted to the stable alternative WAN link without packet loss. The original degrading link remains monitored but is no longer serving high-priority traffic classes. The traffic steering event has been recorded in the audit log with the before and after routing state. The Multi-Link Health Scoreboard has been updated to reflect the new traffic routing configuration.	
Trigger	The Digital Twin Validation Sandbox returns a passed validation result for the proposed routing change.	
Main Success Scenario	**User Actions**	**System Actions**
		1. The system receives the passed validation result and the routing change

	8. The user (Network Administrator) views the updated routing configuration on the Multi-Link Health Scoreboard.	payload from the Digital Twin Sandbox module. 2. The system initiates the hitless handoff sequence by pre-computing the new flow table entries for the target alternative WAN link. 3. The system submits the updated flow table entries to the SDN controller northbound API using the OpenFlow or NETCONF protocol. 4. The system simultaneously maintains all existing active session states to preserve TCP sessions and VoIP call state during the transition. 5. The system instructs the SDN controller to atomically apply the new flow table entries, rerouting high-priority traffic classes to the stable alternative link. 6. The system verifies that the traffic rerouting has taken effect by confirming the updated flow table state through the SDN controller API. 7. The system records the traffic steering event in the audit log, capturing the prediction trigger health score, confidence level, the routing change applied, and the timestamp. 9. The system updates the Multi-Link Health Scoreboard to reflect the new active routing configuration for the affected WAN links. 10. The system continues monitoring the predictive health scores for both the original degrading link and the new active link.
Alternate Scenarios	**User Actions**	**System Actions**
	3.a. The system receives an API error response from the SDN controller when submitting the flow table update.	3.a. The system logs the SDN controller API error in the audit log, retries the flow table update submission up to three times with an exponential backoff interval, and sends an alert to the Network Administrator if all retry

	5.b. The system detects that the target alternative WAN link has also degraded below the health score threshold during the handoff execution.	attempts fail. 5.b. The system aborts the handoff to the initial target link, evaluates all remaining available WAN links by health score, selects the highest-scoring alternative link, and initiates a new Digital Twin validation cycle for the revised routing change.
Additional Notes (Constraints, etc.)	The traffic steering execution must complete the full flow table update submission within 50 milliseconds from receiving the validation result to achieve hitless handoff performance targets.	

11. Appendix B – Use Case 4 – Validate Routing Change

Use Case ID	UC-4
Use Case Name	Validate Routing Change
Brief Description	This use case describes the automated process by which the PathWise AI Digital Twin Validation Sandbox simulates and validates every AI-generated routing change within a Mininet-based virtual environment. The system uses Batfish policy analysis to verify that the proposed change is loop-free and compliant with existing firewall and security policies before any deployment to the live production network.
Frequency of Use	Medium – 10%
Priority	High
Current Version	1.0
Date of First Version	03/29/2026
Date of Last Version	03/29/2026
Created By	Sricharitha Katta
Last Update By	Sricharitha Katta
Approved By	Bharadwaj Jakkula
Assumptions	A current and accurate virtual topology model of the live network exists within the Digital Twin environment. Batfish has been loaded with the current network device configuration files and firewall policy rules. The Mininet simulation environment is operational and capable of instantiating the network topology. The desired item is available for purchase.
Primary Actor	PathWise AI Digital Twin Validation Sandbox (automated)

Secondary Actor/s	Network Administrator	
Preconditions	The Digital Twin Sandbox has received a routing change proposal from the Predictive Telemetry or Traffic Steering module. The Mininet virtual topology mirror of the live network is pre-initialized and ready for simulation. Batfish has loaded the current network policy and configuration baseline.	
Postconditions	The routing change proposal has received either a passed or failed validation result. The validation result, including the loop check result and policy compliance status, has been recorded in the audit log. If the result is passed, the routing change payload has been forwarded to the Autonomous Traffic Steering module for production deployment. If the result is failed, the routing change has been rejected and an alert has been sent to the Network Administrator.	
Trigger	The Predictive Telemetry Engine or Autonomous Traffic Steering module submits a routing change proposal to the Digital Twin Validation Sandbox API.	
Main Success Scenario	**User Actions**	**System Actions**
		1. The system receives the routing change proposal payload from the requesting module, including the proposed flow table entries and affected link identifiers. 2. The system instantiates a Mininet virtual network topology that mirrors the current live network configuration, including all active links and routing policies. 3. The system applies the proposed routing change to the Mininet virtual topology. 4. The system executes a routing loop detection analysis within the Mininet simulation to verify that the proposed change does not introduce any routing loops. 5. The system submits the proposed configuration change to Batfish for network policy compliance analysis. 6. The system receives the Batfish policy compliance report, which includes checks against existing firewall rules, access control lists, and security policies. 7. The system evaluates the combined results of the loop detection analysis and the Batfish policy compliance report. 8. The system records the full validation

	10. The user (Network Administrator) reviews the validation result and audit log entry in the dashboard.	result, including the loop check outcome, policy compliance status, and any identified violations, in the audit log. 9. The system forwards the passed routing change payload to the Autonomous Traffic Steering module if both the loop detection and policy compliance checks have passed.
Alternate Scenarios	**User Actions**	**System Actions**
	4.a. The system detects a routing loop in the proposed change during the Mininet simulation. 6.b. Batfish identifies a firewall policy violation in the proposed routing change.	4.a. The system immediately terminates the validation cycle, records the loop violation details in the audit log, marks the routing change proposal as failed, and sends an alert notification to the Network Administrator with a description of the detected loop. 6.b. The system marks the routing change proposal as failed due to a policy compliance violation, records the specific policy rule that was violated in the audit log, and sends an alert notification to the Network Administrator with the policy violation details.
Additional Notes (Constraints, etc.)	The full Digital Twin validation cycle must complete within 5 seconds from the time the routing change proposal is received to meet the system quality requirements for timely production deployment.	

12. Appendix B – Use Case 5 – Manage Network Policy

Use Case ID	UC-5
Use Case Name	Manage Network Policy
Brief Description	This use case describes the process by which a Generalist IT Staff member or Network Administrator accesses the Intent-Based Management Interface to define, update, or remove traffic management policies using natural language commands. The system translates the administrator intent into valid YANG/NETCONF configurations and deploys them to the integrated SDN controller.
Frequency of Use	Medium – 10%
Priority	High
Current Version	1.0
Date of First Version	03/29/2026

Date of Last Version	03/29/2026	
Created By	Meghana Nalluri	
Last Update By	Meghana Nalluri	
Approved By	Vineeth Reddy Kodakandla	
Assumptions	The user is authenticated and has policy management permissions assigned to their role. The SDN controller northbound API is accessible and responsive. The natural language processing service within the IBN interface is operational.	
Primary Actor	Generalist IT Staff	
Secondary Actor/s	Network Administrator	
Preconditions	The user has valid PathWise AI credentials with policy management role permissions. At least one SDN controller is integrated and connected to the PathWise AI platform. The IBN interface natural language processing service is running.	
Postconditions	The submitted policy has been translated into a YANG/NETCONF configuration and validated against the current network state. The validated policy configuration has been deployed to the SDN controller and is actively enforced on the network. The new policy has been saved to the policy database and is visible in the policy management dashboard. An audit log entry has been created recording the policy change, the submitting user, and the timestamp.	
Trigger	The user navigates to the Intent-Based Management Interface within the PathWise AI dashboard and initiates a policy definition action.	
Main Success Scenario	**User Actions**	**System Actions**
	1. The user navigates to the Policy Management section of the PathWise AI IBN interface.	2. The system displays the IBN policy management interface, including the natural language input field, active policy list, and policy history panel.
	3. The user types a natural language policy command into the policy input field, for example, "Prioritize VoIP traffic over guest Wi-Fi on all sites."	4. The system processes the natural language command using the NLP engine and displays a human-readable summary of the interpreted policy intent for the user to review.
	5. The user reviews the translated policy summary displayed by the system.	6. The system translates the confirmed policy intent into a YANG/NETCONF configuration payload targeted at the

	7. The user confirms the policy submission by clicking the Apply Policy button. 8. The user reviews the confirmation notification and verifies the policy is listed in the active policy panel.	integrated SDN controller.
Alternate Scenarios	**User Actions**	**System Actions**
	4.a. The system cannot interpret the natural language command with sufficient confidence to generate a valid policy translation. 8.b. The SDN controller returns an error when the system attempts to deploy the YANG/NETCONF configuration.	4.a. The system displays an interpretation error message informing the user that the command could not be processed, provides suggestions for rephrasing the command, and invites the user to re-enter the policy intent. 8.b. The system logs the deployment error, rolls back the policy submission, notifies the user that the policy could not be deployed with a description of the SDN controller error, and preserves the previous network policy state.
Additional Notes (Constraints, etc.)	The IBN interface must support all common administrative tasks without requiring any command-line knowledge, in compliance with the Req-Qual-Use-1 quality requirement.	

13. Appendix B – Use Case 6 – Authenticate User

Use Case ID	UC-6
Use Case Name	Authenticate User
Brief Description	This use case describes the process by which any user type accesses the PathWise AI management platform by providing valid credentials, and the system authenticates the user and grants role-based access permissions appropriate to the user's designated role within the organization.
Frequency of Use	Low– 5%
Priority	High
Current Version	1.0
Date of First Version	03/29/2026
Date of Last Version	03/29/2026
Created By	Bharadwaj Jakkula
Last Update By	Bharadwaj Jakkula
Approved By	Sricharitha Katta
Assumptions	The user has been registered in the PathWise AI user management system by a Network Administrator or IT Manager. The user has a valid, active account with a designated role. The system is accessible via the user's web browser.

Primary Actor	Network Administrator / Generalist IT Staff / MSP Technician / IT Manager	
Secondary Actor/s	None.	
Preconditions	The PathWise AI management platform is online and accessible. The user has a registered account with an active status in the user management database. The user has a stable internet connection.	
Postconditions	The user is authenticated and has been granted access to the PathWise AI dashboard with permissions scoped to their designated role. The successful authentication event has been recorded in the security audit log.	
Trigger	The user navigates to the PathWise AI management platform login page and initiates the sign-in process.	
Main Success Scenario	**User Actions**	**System Actions**
	1. The user navigates to the PathWise AI management platform login URL using a web browser.	
		2. The system displays the secure login page with email address and password input fields.
	3. The user enters their registered email address and password into the respective input fields.	
		4. The system validates that both input fields are populated before proceeding.
	5. The user clicks the Sign In button to submit their credentials.	
		6. The system verifies the submitted credentials against the encrypted user database using secure credential comparison. 7. The system retrieves the user's designated role and permission set from the user management database upon successful credential verification.
	8. The user is directed to the main PathWise AI dashboard appropriate for their designated role.	
Alternate Scenarios	**User Actions**	**System Actions**
	5.a. The user submits incorrect credentials. 5.b. The user's account has been locked due to exceeding the maximum number of failed login attempts.	
		6.a. The system increments the failed login attempt counter for the user account, displays a generic authentication failure message without

		revealing whether the email or password was incorrect, and locks the account after five consecutive failed attempts. 6.b. The system displays a message informing the user that their account has been locked and provides instructions for contacting the system administrator to unlock the account.
Additional Notes (Constraints, etc.)	All credential transmission between the user browser and the PathWise AI server must occur exclusively over TLS 1.3 encrypted connections in compliance with Req-Qual-Sec-1.	